Effects of meals and meal times on uptake of lead from the gastrointestinal tract in humans.

Twenty three adults ingested 203Pb as lead acetate on the 12th hour of a 19 h fast. Retention measured 7 days later in a whole-body counter was 61% and whole-body turnover rates suggested that initial uptake had been considerably greater. Balanced meals eaten with 203Pb reduced lead uptake to 4% and the influence of the food lasted for up to 3 h. The effects of phytate, ethylene-diaminetetra acetate (EDTA), caffeine, alcohol, glucose, a liquid meal and a light snack were tested separately with intermediate results. The effect of a meal was probably largely due to its content of calcium and phosphate salts but lead uptake was probably further reduced by phytate which is plentiful in whole cereals and it was probably increased by a factor in milk. Uptake with skimmed milk was the same as with whole milk and we suggested that the factor was not fat. Comestibles with low mineral and phytate contents reduced lead uptake by intermediate amounts, possibly by stimulation of digestive secretions. The avid uptake of lead during a fast, the large reduction of lead uptake with meals and the likelihood of variations in gastric-emptying rates and dietary habits may be major causes of variation in body burdens of lead in the population.